

Experiment no: 2

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Question:

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION PAGE:

Chunking Analysis of the instruction page, it is a cognitive strategy that breaks down information in smaller, manageable units for easier processing.

The Mind Tango instruction page effectively utilizes chunking in the following ways:

Clear instructions provide user friendly experience; The instructions are broken into 5 easy understandable steps. Each step reduces cognitive overload and makes understanding easier. The concept of chunking is used as the images are grouped under animals, aquatic animals etc. Observation phase: Here this stage focuses on keen observing, users will understand what to expect in the upcoming phase. Memorization Strategy: Encourages users to remember items. Visual Hierarchy and Design Elements

Bold, large title ("MIND TANGO") → Grabs attention and clearly states the task.

Bullet points and spacing → Reduce clutter, enhancing readability.

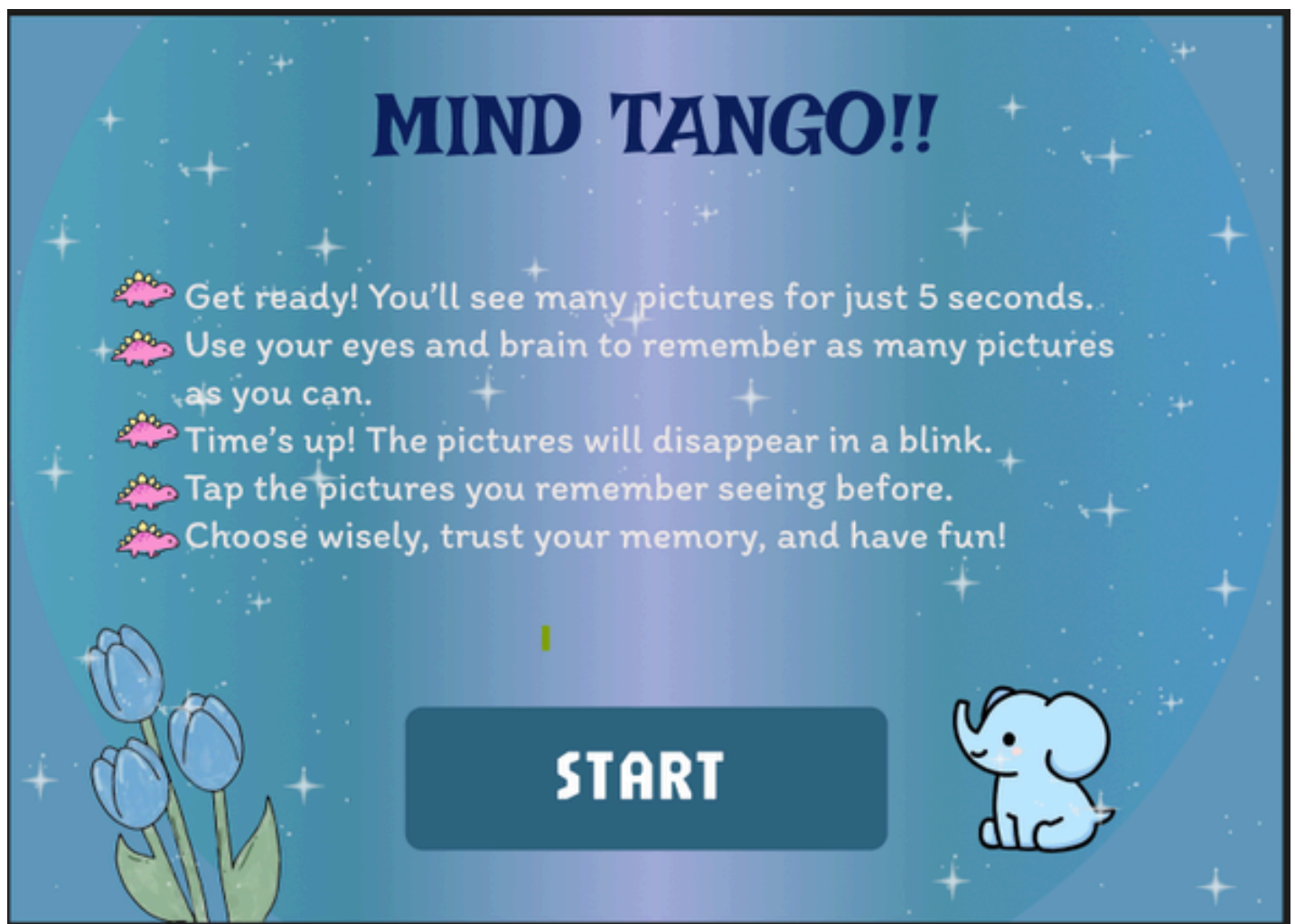
Highlighted "START" button → Signals the next step, keeping navigation intuitive.

Animal-themed visuals → Add a playful touch, engaging users without distraction. 4. Time Constraint Reinforcement

Step 5 explicitly states, "You will have 5 seconds to view the items."

This reinforces expectations while subtly urging users to focus. 5. Simplicity and Clarity

Instructions use short, direct sentences, avoiding unnecessary complexity. The active voice makes it more engaging and action-oriented



FRAME 2:

CHUNKING PHASE:

Analysis of the MIND TANGO - Chunking Phase Screen This screen represents the Chunking Phase of mind tango, where users observe and memorize different icons within a limited time. Below is a breakdown of its key components: 1. Purpose of the Screen

This is the visual memory encoding phase, where users view and group items mentally before recalling them.

The term "Chunking Phase" suggests that the game applies the chunking principle, which helps users remember items by categorizing them into meaningful groups.

Indicates that users have 5 seconds to observe and memorize the displayed items. Various icons (such as seal, hamster, dinosaur, cat, frog, monkey etc.) are displayed in a 5x4 grid format.

These icons are visually distinct yet grouped by similarities, encouraging chunking strategies like: o Categorizing by theme (e.g., animals, aquatic animals). Grouping similar figures or backgrounds (e.g., pink, blue, or yellow tiles). Associating repeated icons (e.g. seal, hamster, dinosaur, cat, frog and monkey). Child friendly animal figures provide mascot, enhancing engagement and giving a playful theme.

The scroll may imply instructions or guidance in later stages

3. How the Chunking Phase Works

1. Users scan the grid and look for patterns or related items to create mental chunks. 2. The countdown timer limits observation time, forcing quick memory strategies. 3. Once time is up, users transition to the recall phase, where they must identify previously seen items. 4. Cognitive and UX Benefits of Chunking. Enhances short-term memory by allowing

users to recall groups of information instead of individual elements. Reduces cognitive overload by helping users organize data efficiently. Improves pattern recognition, making recall easier and more accurate.



FRAME 3:

RECALL PHASE:

Analysis of the MIND TANGO- Selection Phase This screen represents the Selection Phase of a Memory Recall Task, where users recall and choose the items they remember from the previous Chunking Phase. Below is a breakdown of its components: 1. Purpose of the Screen This is the memory retrieval stage, where users select the items, they remember seeing in the previous phase.

The goal is to test the effectiveness of chunking and short-term memory retention.

2. Key Elements and UI Components Title & Instructions "MIND TANGO" (Bold Header) – Reinforces the purpose of the task. A set of many icons are presented as multiple-choice options. Some icons were previously displayed, while others are distractors (new icons added to confuse users). Users must identify which icons appeared in the Chunking Phase. "QUICK OBSERVE" Button (Bottom Center) motivates the users to make decisions fast Once users have made their selections, they press "SUBMIT" to confirm their recall choices.

3. How the Selection Phase Works

1. Users analyze the displayed icons and recall which ones they saw in the Chunking Phase.
2. They select the remembered icons using the radio buttons below each option.

3. Some icons are distractors, testing whether the user's memory is accurate or if they mistakenly recall incorrect icons.
4. Clicking and finalizes their choices, leading to a results or feedback screen.
4. Cognitive and UX Benefits Tests memory accuracy by comparing user selections with previously displayed items. Incorporates distractors to evaluate how well users distinguish real vs. false memories. Gamified elements (animal character, bright colours) make the task engaging.



FRAME 4:

RESULT PAGE:

Analysis of the MIND TANGO- Score & Feedback Screen This screen represents the Score & Feedback Phase of the Memory Recall Task, where users receive their performance evaluation based on the selections made in the previous Recall Phase.

1. Purpose of the Screen Provides feedback on recall accuracy by showing the stars.

Allows users to decide their next action (restart, or quit). 2. Key Elements & UI Components "YAY YOUR SCORE IS!" – Indicates that the user's performance is being displayed.

A star board displays the number of right choices.1"RESTART" – Allows the user to retake the memory recall test. 3. "QUIT" – Ends the task and exits the game. Animated Animal Character (Bottom Left).

Holds a blank sign, possibly for dynamic messages or additional instructions. 3. How This Phase Works 1. The game evaluates the user's selections from the Recall Phase. 2. It calculates the accuracy star score and displays it on a board. 3.

Users review their performance and choose their next action. Restart if they want to improve their recall. Quit if they want to stop playing. 4. Cognitive & UX Benefits

Instant feedback helps users track their memory performance.

Multiple options (Restart,Quit) give users control over their learning experience.

Visual & gamified elements make the task engaging and less stressful.

